

Applying Corollary 3.3 to the L^2 - m -approximable sequence $(W_r^2(\text{PD}(\mathcal{X}_i), \mu_{\mathcal{X}}))_i$ and relying on the fact that $W_r^2(\hat{\mu}_{\mathcal{X}}, \mu_{\mathcal{X}}) \rightarrow 0$ in probability ($m \rightarrow \infty$), verifies (3.1).

The left-hand side of the event in the probability in (3.2) is at most

$$\begin{aligned} & \max_{0 \leq s \leq m^{-1/2}} \frac{1}{\sqrt{m}} \sum_{i=1}^{\lfloor ms \rfloor} \left(W_r(\text{PD}(\mathcal{X}_i), \mu_{\mathcal{X}}) + W_r(\mu_{\mathcal{X}}, \hat{\mu}_{\mathcal{X}}) \right) W_r(\hat{\mu}_{\mathcal{X}}, \mu_{\mathcal{X}}) \\ & + \max_{m^{-1/2} \leq s \leq 1} \frac{1}{\sqrt{m}} \left| \sum_{i=1}^{\lfloor ms \rfloor} W_r^2(\text{PD}(\mathcal{X}_i), \mu_{\mathcal{X}}) - W_r^2(\text{PD}(\mathcal{X}_i), \hat{\mu}_{\mathcal{X}}) \right| \\ & \leq \frac{1}{\sqrt{m}} \sum_{i=1}^{\lfloor \sqrt{m} \rfloor} \left(W_r(\text{PD}(\mathcal{X}_i), \mu_{\mathcal{X}}) + W_r(\mu_{\mathcal{X}}, \hat{\mu}_{\mathcal{X}}) \right) W_r(\hat{\mu}_{\mathcal{X}}, \mu_{\mathcal{X}}) \end{aligned} \quad (3.3)$$

$$+ \max_{\lfloor \sqrt{m} \rfloor \leq k \leq m} \frac{1}{\sqrt{k}} \left| \sum_{i=1}^k W_r^2(\text{PD}(\mathcal{X}_i), \mu_{\mathcal{X}}) - W_r^2(\text{PD}(\mathcal{X}_i), \hat{\mu}_{\mathcal{X}}) \right| \quad (3.4)$$

The term in (3.3) vanishes by the strong law of large numbers from Corollary 3.3 and the assumption that $W_r(\hat{\mu}_{\mathcal{X}}, \mu_{\mathcal{X}}) \rightarrow 0$ in probability for $m \rightarrow \infty$.

The term in (3.4) vanishes by Assumption 1.1. This verifies (3.2). $\square$

Theorem 3.1. *Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be two samples stemming from independent and real-valued stationary processes $(X_t)_t$, resp., $(Y_t)_t$ with mean $\mathbb{E}[X_0]$, resp., $\mathbb{E}[Y_0]$. Furthermore assume that the partial sum processes $\bar{X}_m(s) = \sum_{i=1}^{\lfloor ms \rfloor} X_i$ and $\bar{Y}_n(s) = \sum_{i=1}^{\lfloor ns \rfloor} Y_i$, $s \in [0, 1]$, satisfy a SLLN, i.e., almost surely*

$$\frac{1}{m} \bar{X}_m(1) \rightarrow \mathbb{E}[X_0] \quad \text{and} \quad \frac{1}{n} \bar{Y}_n(1) \rightarrow \mathbb{E}[Y_0] \quad (3.5)$$

as well as an invariance principle in the Skorokhod topology on $D_1 = D([0, 1])$, namely,

$$\begin{aligned} H_m &:= \frac{1}{\sqrt{m}} (\bar{X}_m(\cdot) - \lfloor m \cdot \rfloor \mathbb{E}[X_0]) \Rightarrow \sqrt{\Gamma_X} B_1, \quad \Gamma_X = \sum_{h \in \mathbb{Z}} \text{Cov}(X_0, X_h); \\ J_n &:= \frac{1}{\sqrt{n}} (\bar{Y}_n(\cdot) - \lfloor n \cdot \rfloor \mathbb{E}[Y_0]) \Rightarrow \sqrt{\Gamma_Y} B_2, \quad \Gamma_Y = \sum_{h \in \mathbb{Z}} \text{Cov}(Y_0, Y_h), \end{aligned}$$

where B_1, B_2 are independent standard Brownian motions. Let $D = \mathbb{E}[X_0] - \mathbb{E}[Y_0]$ and define

$$\begin{aligned} \tilde{D}_{m,n}(s) &= s \left(\frac{1}{\lfloor ms \rfloor} \bar{X}_m(s) - \frac{1}{\lfloor ns \rfloor} \bar{Y}_n(s) \right), \quad s \in [0, 1], \\ \tilde{V}_{m,n} &= \left\{ \int_0^1 (\tilde{D}_{m,n}^2(s) - s^2 \tilde{D}_{m,n}^2(1))^2 \nu(ds) \right\}^{1/2}, \end{aligned}$$

where we use the convention $\bar{X}_m(s)/\lfloor ms \rfloor = 0$ for $s < 1/m$. Suppose there is a constant $\tau \in (0, 1)$ such that $\lim_{m,n \rightarrow \infty} m/(m+n) = \tau$. Then (w.r.t. the Euclidean metric on $\mathbb{R}^2$)

$$\sqrt{m+n}(\tilde{D}_{m,n}^2(1) - D^2, \tilde{V}_{m,n}) \Rightarrow \left(\xi B(1), \xi \left\{ \int_0^1 s^2 (B(s) - sB(1))^2 \nu(ds) \right\}^{1/2} \right),$$

where B is a standard Brownian motion and $\xi = 2\sqrt{\frac{\Gamma_X}{\tau} + \frac{\Gamma_Y}{1-\tau}}(\mathbb{E}[X_0] - \mathbb{E}[Y_0])$.

Proof of Theorem 3.1. As in the formulation of the theorem let B, B_1, B_2 denote independent standard Brownian motions. The proof is divided into several main steps.

Step 1. We start with rewriting $\sqrt{m+n}(\tilde{D}_{m,n}^2(1) - D^2, \tilde{V}_{m,n})$ in a more suitable way. Writing out the first term yields

$$\begin{aligned} \sqrt{m+n}(\tilde{D}_{m,n}^2(1) - D^2) &= \sqrt{m+n}(\tilde{D}_{m,n}(1) - D)(\tilde{D}_{m,n}(1) + D) \\ &= \left(\sqrt{\frac{m+n}{m}} H_m(1) - \sqrt{\frac{m+n}{n}} J_n(1) \right) \\ &\quad \times \left(\frac{1}{m} \bar{X}_m(1) - \frac{1}{n} \bar{Y}_n(1) + \mathbb{E}[X_0] - \mathbb{E}[Y_0] \right) =: Z_{m,n}. \end{aligned} \quad (3.6)$$

The second term satisfies

$$\sqrt{m+n} \tilde{V}_{m,n} = \left\{ \int_0^1 (m+n) (\tilde{D}_{m,n}^2(s) - s^2 \tilde{D}_{m,n}^2(1))^2 \nu(ds) \right\}^{1/2} \quad (3.7)$$

and its integrand equals

$$\begin{aligned} &(m+n) (\tilde{D}_{m,n}^2(s) - s^2 \tilde{D}_{m,n}^2(1))^2 \\ &= (m+n) (\tilde{D}_{m,n}^2(s) - s^2 D^2 - s^2 (\tilde{D}_{m,n}^2(1) - D^2))^2 \\ &= \left(\sqrt{m+n} (\tilde{D}_{m,n}(s) - sD) (\tilde{D}_{m,n}(s) + sD) - s^2 \sqrt{m+n} (\tilde{D}_{m,n}^2(1) - D^2) \right)^2 \\ &= (\tilde{Z}_{m,n}(s) - s^2 Z_{m,n})^2. \end{aligned} \quad (3.8)$$

Here we defined the first term in the last curly parentheses by

$$\begin{aligned} \tilde{Z}_{m,n}(s) &:= \sqrt{m+n} (\tilde{D}_{m,n}(s) - sD) (\tilde{D}_{m,n}(s) + sD) \\ &= s^{1-\gamma} \left\{ \frac{ms/\lfloor ms \rfloor}{\sqrt{m/(m+n)}} \frac{\bar{X}_m(s) - \lfloor ms \rfloor \mathbb{E}[X_0]}{\sqrt{m}} - \frac{ns/\lfloor ns \rfloor}{\sqrt{n/(m+n)}} \frac{\bar{Y}_n(s) - \lfloor ns \rfloor \mathbb{E}[Y_0]}{\sqrt{n}} \right\} \\ &\quad \times s^\gamma \left\{ \frac{1}{\lfloor ms \rfloor} \sum_{i=1}^{\lfloor ms \rfloor} X_i + \mathbb{E}[X_0] - \frac{1}{\lfloor ns \rfloor} \sum_{i=1}^{\lfloor ns \rfloor} Y_i - \mathbb{E}[Y_0] \right\} \\ &=: \tilde{Z}_{1,m,n}(s) \times \tilde{Z}_{2,m,n}(s). \end{aligned} \quad (3.9)$$

Step 2. Next, in view of (3.6), let us introduce the functional

$$F_1: D_1 \times D_1 \rightarrow \mathbb{R}, \quad (u, v) \mapsto 2(\mathbb{E}[X_0] - \mathbb{E}[Y_0]) \left(\frac{u(1)}{\sqrt{\tau}} - \frac{v(1)}{\sqrt{1-\tau}} \right).$$

Then, as an immediate consequence of (3.5), we see that

$$\begin{aligned} Z_{m,n} - F_1(H_m, J_n) &= \left\{ \sqrt{\frac{m+n}{m}} \left(\frac{1}{m} \overline{X}_m(1) - \frac{1}{n} \overline{Y}_n(1) + \mathbb{E}[X_0] - \mathbb{E}[Y_0] \right) - \frac{2(\mathbb{E}[X_0] - \mathbb{E}[Y_0])}{\sqrt{\tau}} \right\} H_m(1) \\ &\quad + \left\{ \sqrt{\frac{m+n}{n}} \left(\frac{1}{m} \overline{X}_m(1) - \frac{1}{n} \overline{Y}_n(1) + \mathbb{E}[X_0] - \mathbb{E}[Y_0] \right) - \frac{2(\mathbb{E}[X_0] - \mathbb{E}[Y_0])}{\sqrt{1-\tau}} \right\} J_n(1) \\ &\rightarrow 0 \end{aligned} \tag{3.10}$$

in probability. Since F_1 is continuous with respect to (any) product metric on $D_1 \times D_1$, we also have by the invariance principle

$$\begin{aligned} F_1(H_m, J_m) &\Rightarrow F_1(\sqrt{\Gamma_X} B_1, \sqrt{\Gamma_Y} B_2) \\ &= 2(\mathbb{E}[X_0] - \mathbb{E}[Y_0]) \left(\frac{\sqrt{\Gamma_X} B_1(1)}{\sqrt{\tau}} - \frac{\sqrt{\Gamma_Y} B_2(1)}{\sqrt{1-\tau}} \right) \\ &\stackrel{\mathcal{L}}{=} 2(\mathbb{E}[X_0] - \mathbb{E}[Y_0]) \sqrt{\frac{\Gamma_X}{\tau} + \frac{\Gamma_Y}{1-\tau}} B(1). \end{aligned}$$

Consequently, an application of Slutsky's theorem yields $Z_{m,n} \Rightarrow \xi B(1)$.

Step 3. In order to derive the weak limit of (3.7) we start by considering the integrand first. In view of (3.8) and (3.9), let us introduce the operator

$$G_{1,1}: D_1 \times D_1 \rightarrow D_1, \quad (u, v) \mapsto \left(s^{1-\gamma} \left(\frac{u(s)}{\sqrt{\tau}} - \frac{v(s)}{\sqrt{1-\tau}} \right) \mid s \in [0, 1] \right),$$

and the function

$$g_{1,2}: [0, 1] \rightarrow \mathbb{R}, \quad s \mapsto 2s^\gamma (\mathbb{E}[X_0] - \mathbb{E}[Y_0])$$

for some $\gamma \in (0, 1)$ arbitrary but fixed. Then we define the operator $G_1 := G_{1,1} \cdot g_{1,2}$ and finally set

$$F_2: D_1 \times D_1 \rightarrow D_1, \quad (u, v) \mapsto (G_1(u, v) - (\cdot)^2 F_1(u, v))^2.$$

Observe that F_2 is discontinuous only on $\text{Dis}(F_2) = D_1 \setminus \mathcal{C} \times D_1 \setminus \mathcal{C}$, where $\mathcal{C}$ denotes the set of continuous functions on the interval $[0, 1]$. Since by assumption

$$(H_m, J_n) \Rightarrow (\sqrt{\Gamma_X} B_1, \sqrt{\Gamma_Y} B_2)$$